

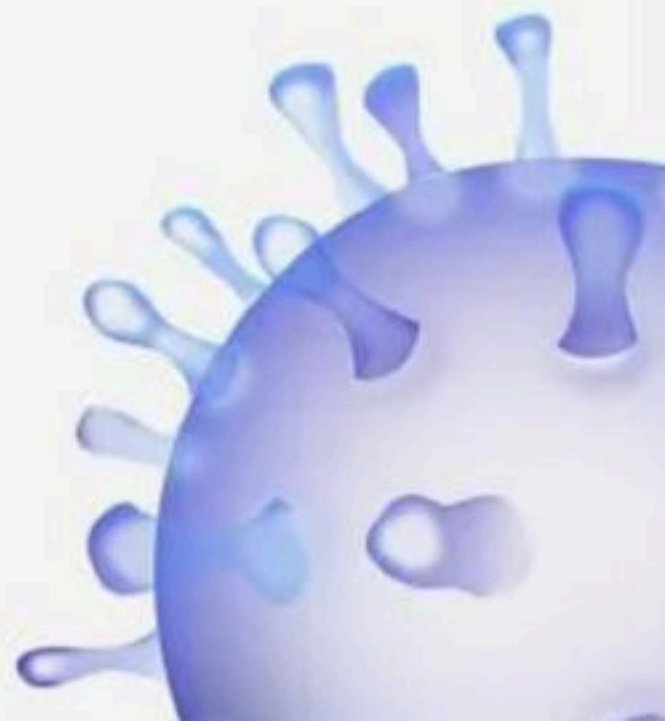
Chemical Coordinate & Integration - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



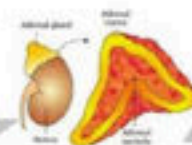
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Chemical Coordination and Integration



BIOLOGYLIVE

6 PM



Chapter - 22 Chemical Coordination and Integration

Points To Remember ✓✓

Endocrine glands : These are ductless glands which secrete hormones directly into the blood stream.

Hormones : Non-nutrient chemicals synthesised in trace amount by Endocrine glands, that act as intracellular messengers and are specific in their action which are transported by blood from site of production to site of action.

* Hypothalamus :

- It is basal part of diencephalon.
- Has neurosecretory cells called nuclei which produce hormones to regulate the synthesis and secretion of pituitary gland hormones.
- Two types of hormones released are :

Releasing hormones : Stimulate secretion of pituitary hormones, *e.g.*, *Gonadotrophin* releasing hormone stimulates pituitary gland to synthesise gonadotrophins.

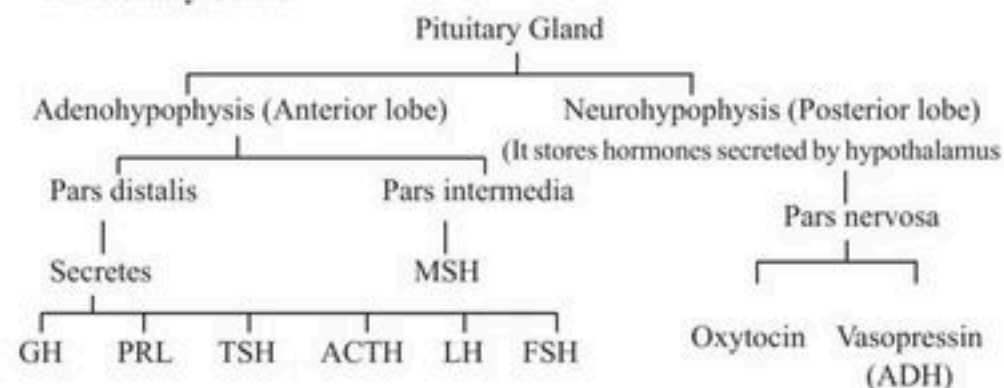
Inhibiting hormones : Inhibit secretions of pituitary hormones, *e.g.*, *Somatostatin* inhibits secretion of growth hormone.

* Pituitary Gland :

- Located in bony cavity called as sella tursica.
- Attached to hypothalamus by a stalk.
- Divided anatomically into : Adenohypophysis (Anterior lobe) and Neurohypophysis (Posterior lobe).
- Hormones released from hypothalamic neurons reach anterior pituitary through portal system and through neurons in Posterior pituitary.
- Posterior pituitary is under neural control of hypothalamus.

8 PM

1. Pituitary Gland



Adenohypophysis : (Anterior lobe of Pituitary)

- **Growth hormone (GH)** : Oversecretion leads to gigantism and low secretion causes dwarfism and Proper secretion leads to proper growth of body.
- **Prolactin (PRL)** : Growth of mammary gland and formation of milk in them.
- **Thyroid stimulating hormone (TSH)** : Stimulates synthesis and secretion of thyroid hormones from thyroid gland.
- **Adrenocorticotrophic hormone (ACTH)** : Stimulates synthesis and secretion of steroid hormones called glucocorticoids from adrenal cortex.
- **Luteinizing hormone (LH)** : Synthesis and secretion of hormones called androgens in males, and helps in ovulation and maintenance of corpus luteum in females.
- **Follicle stimulating hormone (FSH)** : Regulate spermatogenesis in males, and growth and development of ovarian follicles in females.

Neurohypo physis (Posterior lobe of Pituitary)

- **Oxytocin** helps in contraction of uterus during child birth and milk ejection from mammary gland in females.
- **Vasopressin** : Acts on kidney and stimulates reabsorption of water and electrolytes by distal tubules to reduce water loss through urine. It is also called as Anti Diuretic Hormone (ADH).

Diabetes insipidus : Impairment of synthesis of ADH

- Diminished ability of kidney to conserve water.
- Water loss and dehydration.
- Can be overcome by taking more water.

Excessive Secretion of Growth Hormone

- **Acromegaly** : It is a condition when the pituitary gland makes too much growth hormone. It is due to a tumour in pituitary gland. Person suffering from acromegaly (acro means tip and megaly means enlargement) may gradually develop a long face with protruding lower jaw, enlarged nose and wider spacing between teeth and enlarged hands and feet.

2. Pineal Gland :

- Located on dorsal side of forebrain.
- Secretes Melatonin for regulation of 24-hours rhythm, sleep-wake cycle, menstrual cycle, pigmentation etc.

3. Thyroid Gland :

- Has two lobes on either side of trachea interconnected by isthmus (connective tissue).
- Composed of follicles and stromal tissues.
- Follicular cells synthesise thyroxine (T_4) and tri-iodothyronine (T_3).
- Iodine is necessary for normal functioning of thyroid.
- **Goitre (Hypothyroidism)** : Enlargement of thyroid gland; Hypothyroidism may lead to mental retardation and stunted growth (cretinism) Deaf-mutism in the baby if it occurs during pregnancy.
- **Hyperthyroidism** : Occurs due to cancer or due to development of nodules in thyroid glands, affects body physiology as abnormally high levels of thyroid hormones is synthesised. Basic metabolic rate increases.
- **Exophthalmic goitre** : It is a form of hyperthyroidism, characterised by enlargement of thyroid gland, protrusion of eye balls and increased BMR
- Thyroid hormone controls carbohydrates, proteins & fats metabolism.
- Also secretes a protein hormone called Thyrocalcitonin (TCT) which regulates blood calcium level.

4. Parathyroid Gland :

- Present on back side of thyroid gland. Each lobe of thyroid gland has its one pair.
- Secrete peptide hormone called parathyroid hormone (PTH) which increases calcium levels in blood so called **hypercalcemic** hormone.
- PTH stimulates bone resorption, and reabsorption of calcium from blood and reabsorption of calcium by renal tubules, thus increasing blood Ca^{++} level.

5. Thymus Gland

- Located on dorsal side of heart and aorta.
- Secrete peptide hormones called Thymosins which play role in differentiation of T-lymphocytes (help in cell mediated immunity.)
- Thymosins also produce antibodies and provide humoral immunity.
- Immunity of old people usually becomes weak as thymus gets degenerated with age.

6. Adrenal Gland

- Located at anterior part of each kidney.
- Has centrally located adrenal medulla and at periphery in adrenal cortex.
- **Adrenal medulla.**
 1. **Adrenal medulla secretes** adrenaline (epinephrine) and nor adrenaline (norepinephrine), commonly called as catecholamines or emergency hormones or hormones of fight or flight.
 2. These hormones increase heart beat, rate of respiration, breakdown of glycogen thus increase blood glucose level, breakdown of lipids and protein, alertness, raising of hairs, sweating etc.
- **Adrenal Cortex-(3 layers) :**
 - Zona reticularis (inner layer)
 - Zona fasciculata (middle layer)
 - Zona glomerulosa (outer layer)
- **Adrenal cortex secretes :**
 1. **Androgenic steroids :** (Secreted by Zona reticularis)
 - Secreted in small amounts.
 - Play role in growth of axial pubic and facial hair during puberty.
 2. **Glucocorticoids :** (Secreted by fasciculata)
 - Involved in carbohydrate metabolism.
 - Stimulates gluconeogenesis, lipolysis and proteolysis.
 - e.g., Cortisol which is also involved in cardio-vascular and kidney functions.
 - It also suppresses immune response and stimulates RBC production.
 3. **Mineralocorticoids :** (Secreted Zona glomerulosa)
 - Regulate balance of water and electrolytes in body.
 - e.g., Aldosterone which also helps in reabsorption of Na^+ and water. Excretion of K^+ and phosphates ions from renal tubules.

- When adrenal cortex is damaged, it does not produce enough cortisols (which regulate body's reaction to stressful situations) and aldosterone.
- It result in **Addison's disease**. Symptoms of addison's disease are weak muscles, extreme fatigue, increased skin pigmentation, weight loss, sores in mouth and depression.

Two major adrenal insufficiency

- Primary adrenal insufficiency where our immune system mistakes adrenal for an antigen and tries to damage it.
 - Secondary adrenal insufficiency-when pituitary gland can't produce ACTH
7. **Pancreas** : It is called composite/dual gland. As it acts as Exocrine and endocrine gland i.e. has both exocrine and endocrine function.

- Contains about 1-2 million islets of Langerhans which has glucagon secreting α -cells and insulin secreting β -cell.
- **Glucagon** : Peptide hormone, stimulates glycogenolysis by acting on liver cells. Also, stimulates gluconeogenesis. Hence called hyperglycemic hormone.
- **Insulin** : Peptide hormone, acts on hepatocytes and adipocytes to enhance cellular glucose uptake, stimulates conversion of glucose to glycogen (glycogenesis), so decrease blood glucose level. Hence called hypoglycemic hormone.
- Deficiency of insulin causes diabetes mellitus in which loss of glucose occurs through urine. Excessive hunger and thirst (polydipsia) are other symptoms of Diabetes.

- Insulin and glucagon are antagonistic hormones i.e. play opposite role.

Glycogenolysis : Breaking of glycogen into glucose.

Gluconeogenesis : Formation of glucose from substances other than glycogen.

Glycogenesis : Conversion of glucose into glycogen.

8. Testis :

- A pair of testis composed of seminiferous tubules and interstitial cells is present in the scrotal sac of males.
- Leydig cells (interstitial cells) produce androgens (mainly testosterone) which regulate development and maturation of male accessory sex organs, formation of secondary sex characters and play stimulatory role in spermatogenesis. Male sexual behaviour (libido) is influenced by androgens.

9. **Ovary** : A pair of ovaries which produce one ovum in each menstrual cycle are present in abdomen in females.

- Ovary composed of ovarian follicles and stromal tissue.
- Estrogen synthesised by growing ovarian follicles helps in stimulation of growth of female secondary sex organs, female behaviour, mammary gland development and female secondary sex characters.
- Ruptured follicle form corpus luteum which secretes progesterone. Progesterone supports pregnancy and stimulates alveoli formation and milk secretion in mammary glands.

Hormones secreted by tissues which are not endocrine glands :

- (a) **Heart** : Atrial wall secretes Atrial Natriuretic factor (ANF) which decreases blood pressure by dilation of the blood vessels.
- (b) **Kidney** : Juxtaglomerular cells secrete erythropoietin which stimulates erythropoiesis (RBC formation).
- (c) **Gastro-intestinal tract** : It secretes four peptide hormones.
 - **Gastrin** : Acts on gastric glands and stimulates secretion of hydrochloric acid and pepsinogen.
 - **Secretin** : Acts on pancreas and stimulates secretion of water and bicarbonate ion.
 - **Cholecystokinin (CCK)** : Act on pancreas and gall bladder to stimulate secretion of pancreatic juice and bile juice respectively.

Gastric inhibitory peptide (GIP) : Inhibits gastric secretion and motility.

Mechanism of hormone action : By hormone receptors of two kinds, *i.e.*,

(a) **Located on membrane of target cell**

- These are membrane bound receptors.
- Form hormone receptor complex.



Leads to biochemical changes in tissue.



Release of second messengers like (cyclic AMP, IP_3 , Ca^{2+} etc.) which regulate cellular metabolism.



(b) Located inside the target cell

- These are intra cellular receptors.
- Hormones (steroid hormones iodothyronines etc.) interact with them and cause physiological and developmental effects of regulating gene expression.

Questions



(SRT) Select Response Type Question (1 mark each)

1. Which to system coordinate and regulate physiological function of our body?
 - (a) muscular system and endocrine system
 - (b) neural system and muscular system
 - (c) neural system, and endocrine system
 - (d) endocrine system and reproductive system
2. What is the role of melanocyte stimulating hormone?
 - (a) act on hepatic cells
 - (b) act on melanocytes and regulate pigmentation of skin
 - (c) act on ovarian follicles
 - (d) act on osteocytes
3. Name the hormone which act antagonistically in order to regulate calcium level in blood?
 - (a) Glucocorticoid and adrenocortical
 - (b) Thyrocalcitonin and parathyroid hormone
 - (c) Mineralocorticoid and Thyrocalcitonin
 - (d) Parathyroid hormone and glucocorticoid

CONSTRUCTED RESPONSE TYPE (CRT)

Very Short Answer Questions (1 mark each)

4. Give the names of any one glucocorticoid and one mineralocorticoid.
5. How does Atrial natriuretic factor decrease blood pressure ?

6. Which structure is formed from ruptured follicle in females ? What is its role ?
7. Immunity of old persons becomes very weak. Give reason.

Short Answer Questions-I

(2 marks each)

8. What happens if a person suffers from prolonged hyperglycemia ?
9. What are the two modes through which the hypothalamus causes the release of hormones by pituitary gland ?
10. Androgen regulates the development, maturation and other important functions in human male. List them.
11. Mr. Akshay notices that his shoe size has progressively increased. He also observes that shape of his face is gradually changing with protruding lower jaw. What can be the cause for all changes. Name the disorder.

Short Answer Questions-II

(3 marks each)

12. Define hormone and classify them on basis of their chemical nature.
13. How do oxytocin, progesterone and estrogen differ from each other ?
14. What are the disorders caused and the effects produced due to malfunctioning/ improper secretion from thyroid gland ?
15. Name the disease/disorder caused by :
 - (a) Excessive secretion of Thyroid hormone in adults.
 - (b) Insufficient amount of insulin secreted by pancreas.
 - (c) Damage of adrenal cortex.

Long Answer Questions

(5 mark each)

16. 'The master gland regulates a number of physiological functions in our body.' Give reasons and explain.



Answers

(SRT) Select Response Type Question (1 mark each)

1. (c) Nervous system and endocrine system.
2. (b) Acts on melanocytes and regulates pigmentation of skin.
3. (b) Thyrocalcitonin (TCT) and parathyroid hormone (PTH).

Very Short Answers (1 mark each)

4. Glucocorticoid—Cortisol; Mineralocorticoid—aldosterone.
5. By dilation of the blood vessels.
6. Corpus luteum which secretes progesterone.
7. Thymus gland degenerates with age.

Short Answer Questions-I (2 marks each)

8. Gets affected by diabetes mellitus which causes loss of glucose through urine and formation of harmful ketone bodies.
9. Through hypothalamic neurons control anterior pituitary gland. Through neural regulation controls posterior pituitary gland.
10. Refer Points to Remember.
11. Increased secretion of growth hormone Acromegaly

Short Answers Questions-II (3 marks each)

12. Refer Points to Remember of this support material
13. Oxytocin causes milk ejection and contraction of uterus at time of child birth.
Progesterone—causes milk secretion and maintains pregnancy.
Estrogen : Refer Points to Remember.
14. Refer Points to Remember.
15. (a) Exophthalmic goitre (b) Diabetes
(c) Addison's disease

Long Answers Questions-II**(3 mark each)**

16. Explain the role of pituitary gland + Refer Points to Remember.

***Case Study based question :** CBQs

17. Read the following and answer any four questions from 17 (i) to 17 (v) given below :

Xenoestrogens are found in a variety of everyday items. Its sources are plastic, pesticides, insecticides, beauty products etc. Many of us don't think twice about the make up we wear, container we use to pack a lunch. Unfortunately this may be altering the way our body works naturally because they all contain endocrine disruptors called Xenoestrogens. Xenoestrogens are a subcategory of endocrine disruptor that specifically has estrogen like effects. Estrogen is a natural hormone synthesized by the growing ovarian follicles in females. It helps in the growth of female secondary sex organs and female secondary sex characters. It is also important for bone growth and reproduction in men and women. When xenoestrogen enter the body they increase the total amount of estrogen in the body resulting in a phenomenon called estrogen dominance. As they are not biodegradable they are stored in our fat cells. This leads to breast cancer, prostate cancer, obesity, infertility, miscarriages and diabetes.

(i) Xenoestrogen mimics the action of _____ and its excess causes _____.

- (a) Progesterone, bone and prostate cancer
- (b) Estrogen, breast and prostate cancer
- (c) Cortisol, brain and blood cancer
- (d) Thyrocalcitonin, bone and lung cancer

(ii) Estrogen is secreted mainly by :

- (a) GI tract
- (b) Pancreas

- (c) Growing ovarian follicles
- (d) Thymus
- (iii) Select the function/functions performed by estrogen in females :
 - (a) Growth of female secondary sex organs
 - (b) Bone growth
 - (c) Female secondary sex-characters
 - (d) All of these
- (iv) What measures, as an individual you would take to reduce your exposure to these harmful exoestrogens?
 - (a) Eat local and organic food
 - (b) Use beauty products made from natural ingredients
 - (c) Do not use plastic lunch box and water bottles
 - (d) All of these

✓ **(v) Assertion :** Estrogen belongs to the class of steroid hormones.

Reason : Cholesterol is the precursor used in synthesizing estrogen hormone.

- (a) Both assertion & reason are true, and the reason is the correct explanation of the assertion.
- (b) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.
- (c) Assertion is true but reason is false
- (d) Both the assertion and reason are false

(SRT) Assertion and Reason type Questions-II : (1 mark each)

In each of the following questions, two statements are given, one is Assertion and other is Reason. Mark the correct answer as :

- (a) Both assertion & reason are true, and the reason is the correct

explanation of the assertion.

(b) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.

(c) Assertion is true but reason is false

(d) Both the assertion and reason are false

18. **Assertion :** Hormone thyrocalcitonin has antagonistic effect to that of parathyroid hormone (PTH).

Reason : Thyrocalcitonin decreases the blood calcium level while parathyroid hormone increases blood calcium level.

19. **Assertion :** Immunity of old person becomes weak.

Reason : Thymus degenerates in old individuals.



Answers :

17. (i) (b) Estrogen, breast and prostate cancer

(ii) (c) Growing ovarian follicles

(iii) (d) All of these

(iv) (d) All of these

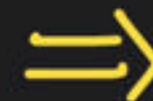
(v) (a) Both assertion & reason are true, and the reason is the correct explanation of the assertion

18. (a) Both assertion & reason are true, and the reason is the correct explanation of the assertion.

19. (a) Both assertion & reason are true, and the reason is the correct explanation of the assertion.

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Title: Chemical Coordination and Integration

1. A boy after attaining sexual maturity shows muscular growth, growth of facial and axillary hair, aggressive and low pitch or voice. These changes are attributed to ____ hormone

(1) Estrogen

✓(2) Testosterone ✓✓

(3) Secretin

(4) Glucagon



Title: Chemical Coordination and Integration

2. Statement A : The thymus gland is a lobular structure located on the ventral side of the heart and the aorta. ✓

Statement B : Thymosins play a major role in the differentiation of ^T~~X~~-lymphocytes, which provide cell-mediated immunity. x

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

3. Statement A : Glucocorticoids, particularly cortisol, ^{anti} produces inflammatory reactions and ^{Suppress} stimulates the immune response.

Statement B : Aldosterone helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.

(1) Statement A is correct but Statement B is incorrect.

✓ (2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

4. Mark the correct statement regarding somatostatin.

- (1) It is secreted from anterior pituitary.
- ✓ (2) It inhibits the release of growth hormone.
- (3) It is secreted from posterior pituitary.
- (4) It stimulates STH secretion.



Title: Chemical Coordination and Integration

5. Statement A : Hypothalamus is the basal part of diencephalon, forebrain and it regulates a wide spectrum of body functions.

Statement B : Hypothalamus contains several groups of neurosecretory cells called nuclei which produce hormones.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

6. Match the hormones listed under Column I with their functions listed under Column II. Choose the answer which gives the correct combination of the alphabets of the two columns.

Column I		Column II	
A	Oxytocin	P	Stimulates ovulation
B	Prolactin	Q	Implantation & maintenance of pregnancy
C	Luteinising hormone	R	Lactation after child birth
D	Progesterone	S	Uterine contraction during labour
		T	Reabsorption of water by Nephrons

- ✓ (1) A – S; B – R; C – P; D – Q
(2) A – S; B – P; C – S; D – R
(3) A – S; B – Q; C – R; D – T
(4) A – T; B – R; C – P; D – S



7. Which of the following organs in mammals does not consist of a central 'medullary' region surrounded by a cortical region?

(1) Ovary

(2) Adrenal

✓✓ (3) Liver ✓✓

(4) Kidney



Title: Chemical Coordination and Integration

8. Statement A : In females, LH induces "ovulation" of fully mature follicles and maintains the corpus luteum.

Statement B : FSH stimulates growth and development of the ovarian follicles in females.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

9. Assertion : Insulin act as a hypoglycemic hormone.

Reason : Deficiency of insulin produces polyuria condition.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

✓ (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and reason are incorrect.



10. In human adult females, oxytocin

- (1) Is secreted by anterior pituitary
- (2) Stimulates the growth of mammary glands
- (3) Stimulates pituitary to secrete vasopressin
- ✓ (4) Causes strong uterine contractions during parturitions



Title: Chemical Coordination and Integration

11. Leydig cells produce a group of hormones called

- ✓ (1) Androgens ↪
- (2) Estrogens
- (3) Aldosterone
- (4) Gonadotropins



12. Which hormone receptors are present on membrane of target cell?

(1) Thyroxin

(2) Oestrogen

✓(3) Insulin → peptide Hormone → membr.

(4) All



13. Posterior pituitary gland is associated with which
hormones?

(1) Oxytocin ✓

(2) Vasopressin ✓

(3) ADH ✓

✓ (4) All are correct



Title: Chemical Coordination and Integration

14. Which of the following pairs of hormones are the examples of those that can easily pass through the cellmembrane of the target cell and bind to a receptor inside it (mostly in the nucleus)? steroid

- (1) Insulin and glucagon
- (2) Thyroxin and insulin
- (3) Somatostatin and oxytocin
- (4) ☒ Cortisol and testosterone



15. Hypersecretion of growth hormones during adulthood causes

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(1) Gigantism

✓✓(2) Acromegaly

(3) Acromicria

(4) None of them



Title: Chemical Coordination and Integration

16. A girl after attaining sexual maturity shows development of growing ovarian follicles, development of mammary glands and high pitch of voice.

These changes are attributed to ____ hormones

(1) Melatonin

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✓ (2) Estrogens

(3) Progesterone

(4) Androgens



Title: Chemical Coordination and Integration

17. When both ovaries of rat are removed then which hormone is decreased in blood : --

(1) Oxytocin

(2) Prolactin

✓ (3) Estrogen

(4) Gonadotrophic releasing factor



18. Over-secretion of GH (growth hormone) in child leads to

2025

(1) Dwarfism

(2) Cretinism

✓ (3) Gigantism

(4) Acromegaly



19. The adrenal medulla secretes hormones These are
commonly known as

(1) Steroids

(2) Terpenes

✓(3) Catecholamine

(4) Hormoamines



Title: Chemical Coordination and Integration

20. Glucagon acts mainly on the, ^{Hepatocyte} _____ insulin acts mainly on _____

2023

- (1) Adipocytes, hepatocytes
- (2) Adipocytes, adipocytes
- (3) Hepatocytes, adipocytes
- (4) ☒ Hepatocytes, hepatocytes and adipocytes



21. Immune response of old persons are weak because

(1) Thymus is degenerated in old individual

(2) Thymosin production decreases

✓ (3) 1 and 2

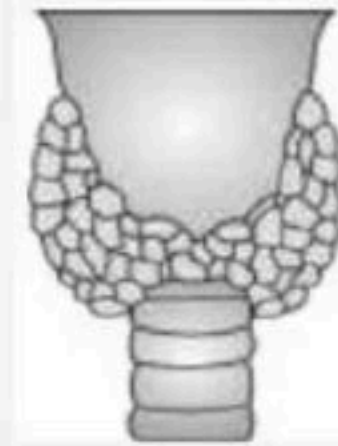
(4) Thymosin production increases



Title: Chemical Coordination and Integration

22. The following figure shows view of the position of thyroid gland. Identify the location of gland.

- (1) Dorsal side of forebrain
- (2) Located on dorsal side of trachea
- (3) Located on the anterior part of each kidney
- ✓ (4) Located on either side of the trachea



Ventral View



Title: Chemical Coordination and Integration

23. Match the columns I and II and choose the correct option.

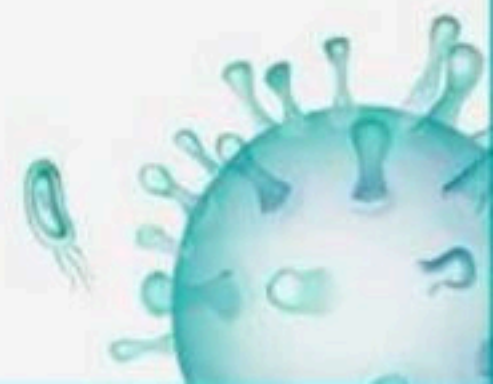
Column I		Column II	
A	Endocrine Glands	P	Produces large number of hormones
B	Hormones	Q	Ductless glands
C	Invertebrates	R	Very simple endocrine system with few hormones
D	Vertebrates	S	Non-nutrient chemical & intercellular messenger

(1) A - P, B - R, C - Q, D - S

(2) A - Q, B - P, C - S, D - R

✓ (3) A - Q, B - S, C - R, D - P

(4) A - P, B - Q, C - R, D - S



Title: Chemical Coordination and Integration

24. After ovulation, the ruptured Graafian follicle transforms into a corpus luteum which mainly secretes

(1) Oestrogen which stimulates the formation of mammary alveoli

✓(2) Progesterone which maintains the endometrium

(3) Relaxin which stimulates the contraction of the myometrium

(4) Progesterone which stimulates the secretion of LH



25. Which of the following hormone is not produced by hypothalamus?

- (1) Somatostatin ✓
- (2) Vasopressin ✓
- (3) Oxytocin ✓
- ✓ (4) All are produced



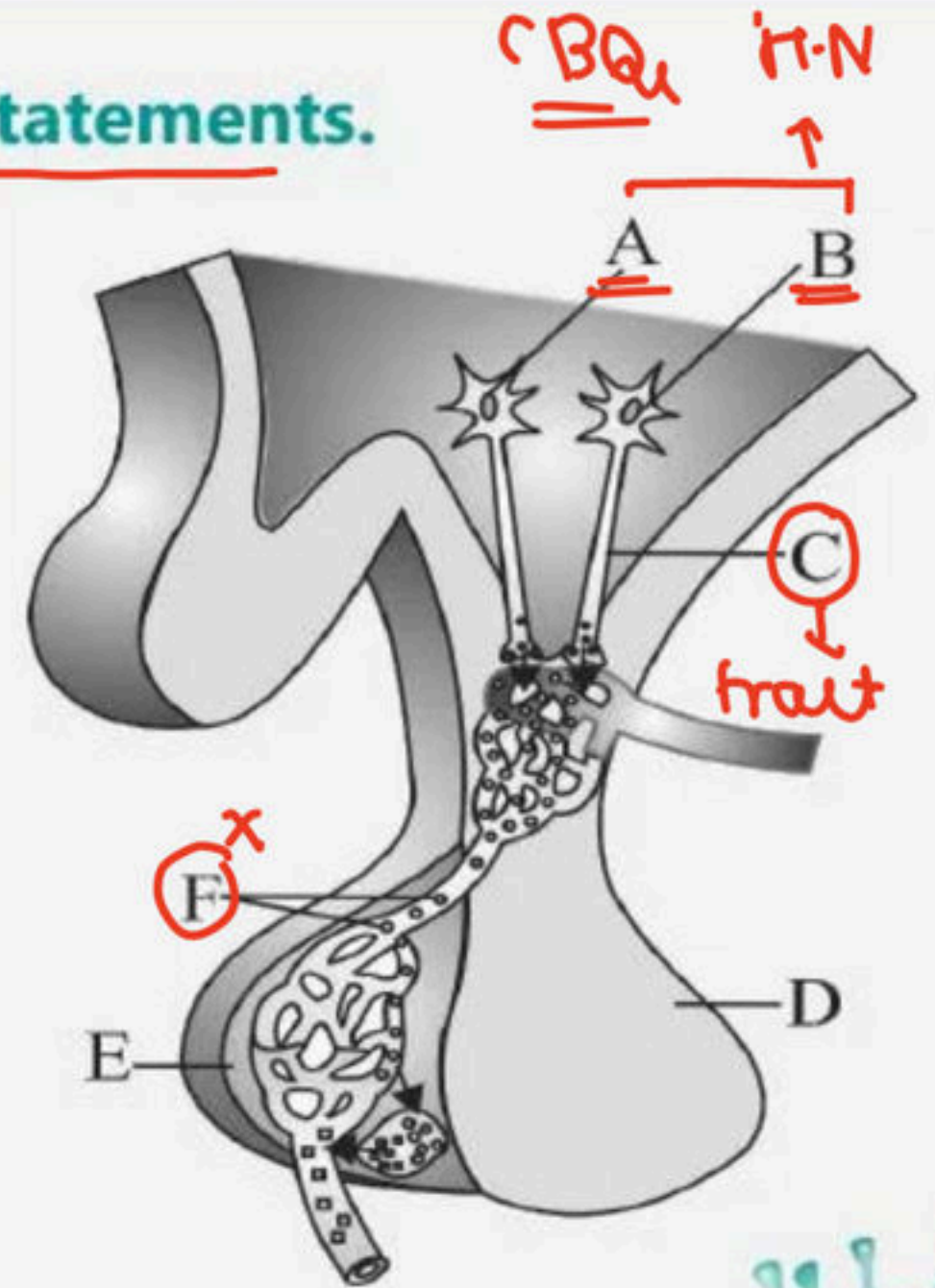
Title: Chemical Coordination and Integration

26. Study the following diagram and select the correct statements.

- ✓ I. 'A' and 'B' are neurosecretory cells.
- II. Oxytocin and ADH are transported through 'F'.
- III. Releasing and inhibiting hormones are transported through 'C'.
- ✓ IV. 'D' is under direct neural regulation of the hypothalamus.

The correct statements are given below:

- ✓ (1) I and IV
- (2) I and II
- (3) III and IV
- (4) II and III



27. Insulin hormone

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- (1) Mainly acts on hepatocytes and adipocytes
- (2) Enhances cellular glucose uptake and utilisation
- (3) Stimulates conversion of glucose to glycogen in the target cells
- (4) All of them



Title: Chemical Coordination and Integration

28. The outer layer of adrenal cortex and inner layer of adrenal medulla are ____ and ____ respectively

2025

- (1) Zona glomerulosa, zona reticularis
- (2) Zona reticularis, zona glomerulosa
- (3) Zona fasciculata, zona reticularis
- ✓ (4) Zona glomerulosa, Nothing



Title: Chemical Coordination and Integration

29. Calcium homeostasis is maintained by

A. Thyroxine

✓ B. Calcitonin

✓✓ C. Parathormone

D. Aldosterone ✗

(1) A and B

✓ (2) B and C

(3) C and D

(4) A and D



30. Aldosterone causes ____ tubule. from renal

(1) Reabsorption of Na⁺ and water

(2) Excretion of K⁺

(3) Excretion of PO₄²⁻ ion

✓ (4) All are correct



31. Insulin and glucagon have :-

(1) Synergistic action

✓(2) Antagonistic action

(3) Similar action

(4) None of them



32. Which one of the following is a second messenger in hormone action?

2025

- (1) Calcium
- (2) IP_3
- (3) cAMP
- ☒ (4) All are correct



33. Which of the following is an incorrect match?

- ✓ (1) ACTH - ^{Adreno}~~Anti~~ - corticotrophic hormone
- (2) LH - Responsible for corpus luteum
- (3) FSH - Promotes spermatogenesis
- (4) All are correct



34. The function of thyroid gland is to :-

- (1) Regulate the basal metabolic rate ✓
- (2) Controls metabolism of fats, proteins ✓
- (3) Regulate the blood calcium level ✓
- ✓ (4) All of the above



35. The glucose homeostasis in blood is thus maintained jointly by the two -

(1) TSH and insulin

(2) Insulin alone

✓(3) Insulin and glucagons

(4) None of these



36. The symptoms of a disease include excretion of large amount of urine and increased thirst

- (1) Hypersecretion of vasopressin- Diabetes mellitus
- (2) Hypersecretion of vasopressin- Diabetes insipidus
- (3) Hyposecretion of vasopressin- Diabetes mellitus
- ✓ (4) Hyposecretion of vasopressin- Diabetes insipidus ✓

ADH Def



37. Fall in levels of oestrogen and progesterone in blood stimulate the production of :-

- (1) GH
- (2) Prolactin
- (3) TSH
- ✓(4) GnRH



38. Mark the mismatch :-

(1) Peptide- Pituitary hormones

✓ (2) Androgen- ^{steroid} ~~Peptide~~ hormones

(3) Amine hormone - Thyroxine derivative

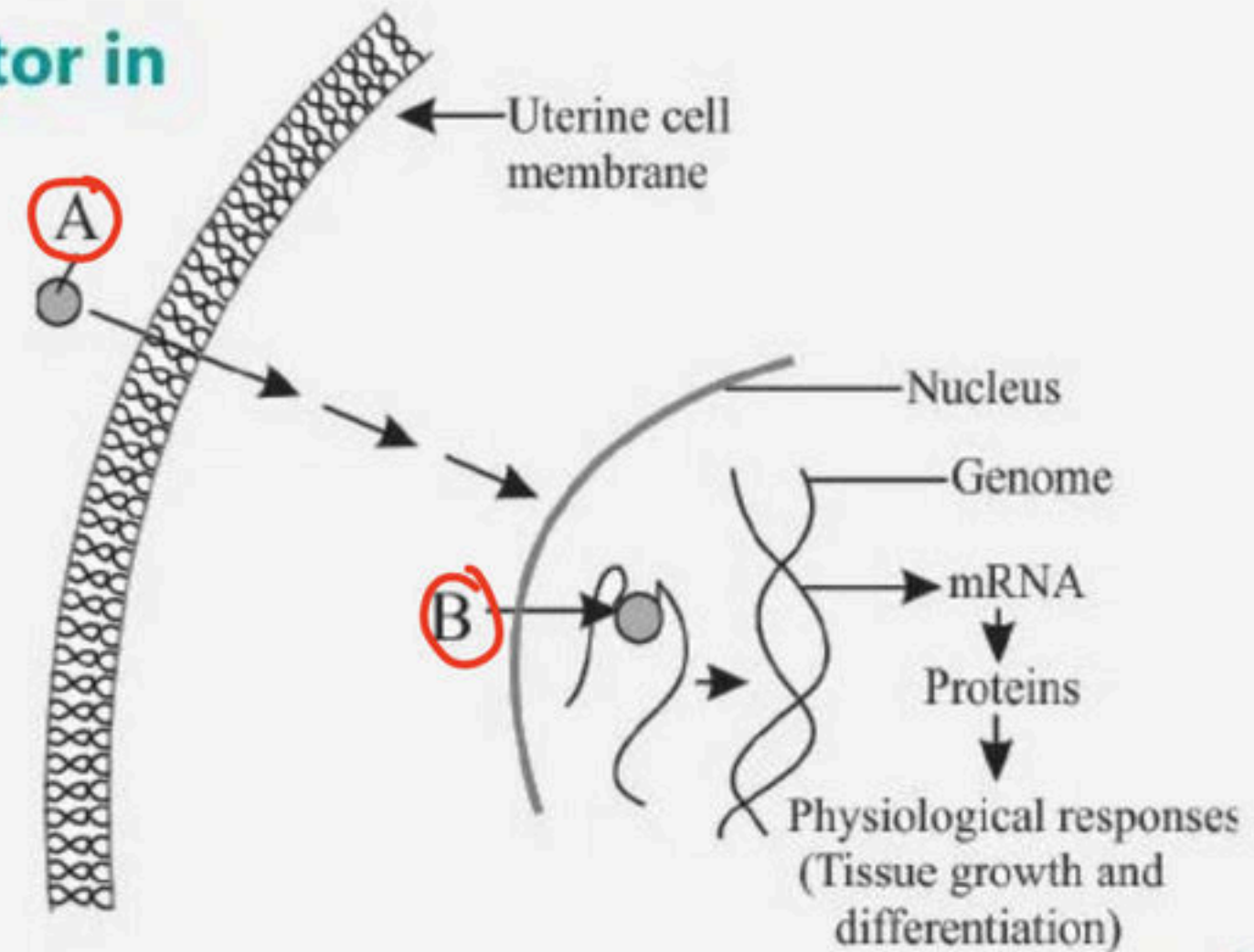
(4) None are correct



Title: Chemical Coordination and Integration

39. Following is the diagram of mechanism of steroid hormone. Identify type of hormone and receptor in given diagram.

- (1) A- FSH, B- Membrane bound receptor
- (2) A- Estrogen, B- membrane bound receptor
- ✓ (3) A- Estrogen, B- Nuclear receptor
- (4) A- FSH, B- Nuclear receptor



40. The heartbeat increases at the time of results of the NEET exams due to

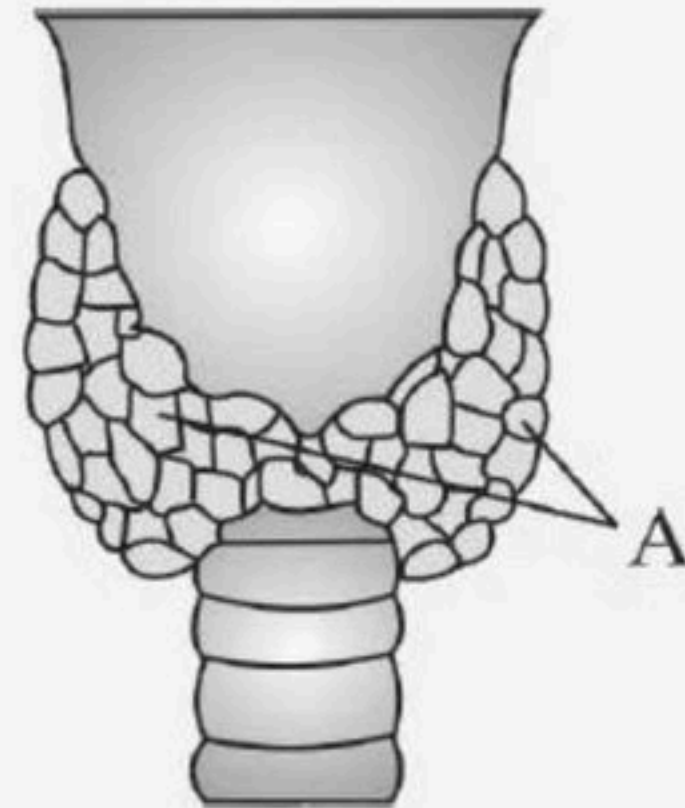
- ✓ (1) Secretion of adrenaline and noradrenaline
- (2) Secretion of adrenaline and corticotropic hormone
- (3) Secretion of adrenaline and renin
- (4) Secretion of adrenaline and cortisol



Title: Chemical Coordination and Integration

41. Which of the following disease is caused due to over secretion of the structure marked as A?

- ✓(1) Grave's disease
- (2) Diabetic mellitus
- (3) Diabetic Insipidus
- (4) Gigantism



Title: Chemical Coordination and Integration

42. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Protein hormone	P	Thyroid hormones
B	Steroids hormone	Q	Cortisol and estradiol
C	Iodothyronines hormone	R	Epinephrine and nor-epinephrine
D	Amino acid derivatives hormone	S	Insulin and Glucagon

CAM+

- (1) A - Q, B - S, C - P, D - R
- (2) A - Q, B - S, C - R, D - P
- ☒ (3) A - S, B - Q, C - P, D - R
- (4) A - P, B - R, C - S, D - Q



43. Ovaries act as the :-

(1) Primary sex organ

(2) Endocrine gland

✓(3) 1 and 2 ✓

(4) Exocrine gland



44. Thymosin

- (1) Amino acid -hormone
- (2) Secreted by pituitary
- (3) Associated with auto immunity
- (4) Promote production of antibodies



Title: Chemical Coordination and Integration

45. Which of the following are correct for progesterone?

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- ✓ (1) It supports pregnancy.
- (2) It regulate female sexual behaviour = *estrogen*
- (3) 1 and 2
- (4) Development of male aggression



46. Hormones are :-

- ✓(1) They are non-nutrient chemicals and are produced in trace amounts
- (2) They are nutrient chemicals and are produced in trace amounts
- (3) They are nutrient chemicals and are produced in huge amounts
- (4) They are non-nutrient chemicals and are produced in huge amounts



Title: Chemical Coordination and Integration

47. Which of the following are NOT under the control of thyroid hormone?

A. Maintenance of water and electrolyte balance ✓

B. Regulation of basal metabolic rate ✓

C. Normal rhythm of sleep-wake cycle ✗

D. Development of immune system ✗

E. Support the process of RBCs formation ✓

Choose the correct answer from the options given below:

(1) A and D only

(2) B and C only

✓ (3) C and D only

(4) D and E only



Title: Chemical Coordination and Integration

48. Hormones are called chemical signals that stimulate specific target tissues. Which is the correct location of these receptors in case of protein hormones?

(1) Extra cellular matrix

(2) Blood

✓(3) Plasma membrane ✓

(4) Nucleus



Title: Chemical Coordination and Integration

49. ____ stimulate growth and development of ovarian follicles in females.

✓ (1) FSH

(2) LH

(3) PRL

(4) TSH



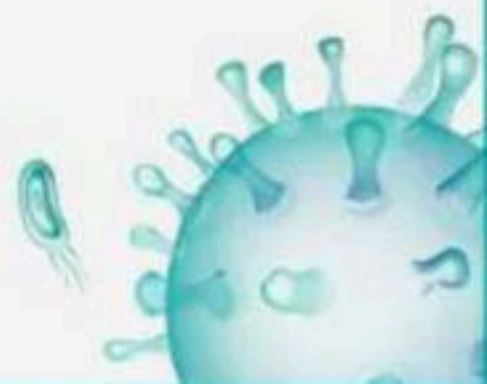
50. Select the incorrect matching.

- ✓ (1) Thyroxine - decreases RBC production, causes inflammation
- (2) α -cell - Glucagon
- (3) β -cell - Insulin
- (4) None are incorrect



51. Which of the following hormone stimulates the synthesis and secretion of glucocorticoids?

- (1) CTH -Cortex Stimulating Hormone
- ✓ (2) ACTH - adrenocorticotrophic hormone
- (3) ACTH -adenocorticotrophic hormone ✗
- (4) GSH - glucocorticois stimulating hormone



Title: Chemical Coordination and Integration

52. Assertion : Insulin is not given orally.

Reason : Insulin cannot be digested in the gut.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

✓ (3) Assertion is correct but Reason is incorrect. ✓

(4) Both Assertion and reason are incorrect.



Title: Chemical Coordination and Integration

53. Statement A : The thyroid gland is composed of two lobes which are located on either side of the trachea.

Statement B : Both the lobes are interconnected with a thin flap of connective tissues called follicles. X

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



54. The hormone ANF is secreted by :-

(1) Graafian follicle

✓ (2) Atrial wall of heart

(3) 1 and 2

(4) None of the above



Title: Chemical Coordination and Integration

55. Which of the given hormones is/are secreted by pineal glands and pars intermedia?

i. FSH

✓ ii. Melatonin

✓ iii. MSH

iv. PRL

v. ACTH

✓ (1) ii and iii

(2) i and ii

(3) ii and iv

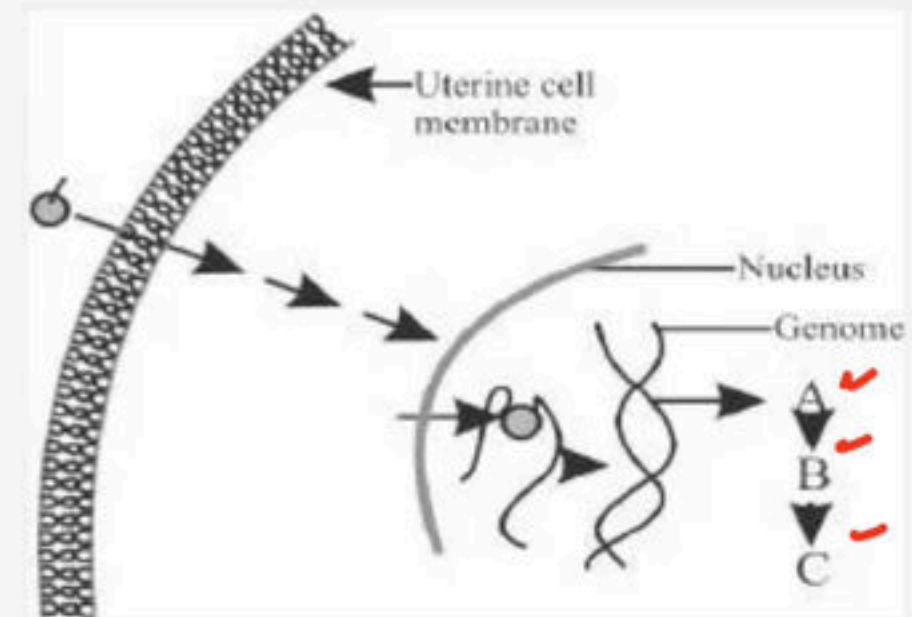
(4) iii, iv and v



Title: Chemical Coordination and Integration

56. Following is the diagram of mechanism of steroid hormone. Identify A, B and C in given diagram.

- (1) A- Tissue growth and differentiation, B- Proteins, CmRNA
- (2) A- Proteins, B- mRNA, C- Tissue growth and differentiation
- (3) A-mRNA, B-Tissue growth and differentiation, CProteins
- ✓(4) A-mRNA, B-Proteins, C- Tissue growth and differentiation



Title: Chemical Coordination and Integration

57. Find the total number of hormones from the following which binds to intracellular receptors.

⁽¹⁾ Cortisol, ⁽²⁾ Testosterone, ⁽³⁾ ACTH, ⁽⁴⁾ Androgenic steroids ⁽⁵⁾ T3, ⁽⁶⁾ Glucagon, Oxytocin, FSH, Progesterone, LH, Oestrogen, GH

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(1) 4

(2) 5

☒ (3) 6

(4) 7



Title: Chemical Coordination and Integration

58. Match Column I with Column II :

Choose the correct answer from the options given below :

Column I		Column II	
A	Exophthalmic goitre	P	Excess secretion of cortisol
B	Acromegaly	Q	Hypo secretion of thyroid hormone and stunted growth
C	Cushing's syndrome	R	Hyper secretion of thyroid hormone & protruding eye balls.
D	Cretinism	S	Excessive secretion of growth hormone

(1) A – P, B – R, C – Q, D – S

(2) A – S, B – Q, C – P, D – R

(3) A – R, B – S, C – Q, D – P

☒ (4) A – R, B – S, C – P, D – Q



Title: Chemical Coordination and Integration

59. A. Increase alertness

B. Sweating

C. Piloerection

D. Increase heart rate

E. Increase respiratory rate

F. Pupillary dilation

Which of the above are effects of emergency hormone?

(1) All except C

(2) All except B and D

(3) All except A

(4) ☒ All are correct



Title: Chemical Coordination and Integration

60. Match the following hormones with the respective disease. Select the correct option.

Column I		Column II	
A	Insulin	P	Addison's disease
B	Thyroxin	Q	Diabetes insipidus
C	Corticoids	R	Acromegaly
D	Growth hormone	S	Goitre
		T	Diabetes <u>mellitus</u>

= Any

(1) A - T, B - P, C - Q, D - R

(2) A - Q, B - S, C - R, D - P

☒ (3) A - T, B - S, C - P, D - R

(4) A - Q, B - S, C - P, D - R



Title: Chemical Coordination and Integration

61. The term 'Medulla' is used in

- (1) Cerebrum ✓✓
- (2) Kidney ✓✓
- (3) Adrenal gland, ovaries ✓✓
- (4) All of them ✓✓



Title: Chemical Coordination and Integration

62. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Hormone receptors	P	Present in the nucleus of target cells
B	Membrane bound receptors	Q	Located in <u>target tissues</u> only
C	Nuclear receptors	R	Present on the cell membrane of the target cells

(1) A - P, B - R, C - Q

☒ (2) A - Q, B - R, C - P

(3) A - R, B - P, C - Q

(4) A - Q, B - P, C - R



Title: Chemical Coordination and Integration

63. Which of the following hormones do not act by a second messenger system?

(1) GH and FSH

(2) Epinephrine

(3) Insulin

(4) Oestrogen



Title: Chemical Coordination and Integration

64. Pineal gland secretes ____ hormone.

(1) MSH

✓ (2) Melatonin

(3) FSH

(4) PSH hormone.



Title: Chemical Coordination and Integration

65. Statement A : Adrenal gland is composed of two types of tissues, centrally located tissue is called adrenal medulla and outside tissue is called adrenal cortex.

Statement B : Adrenal cortex is divided into two layers, called zona reticularis and zona fasciculate.

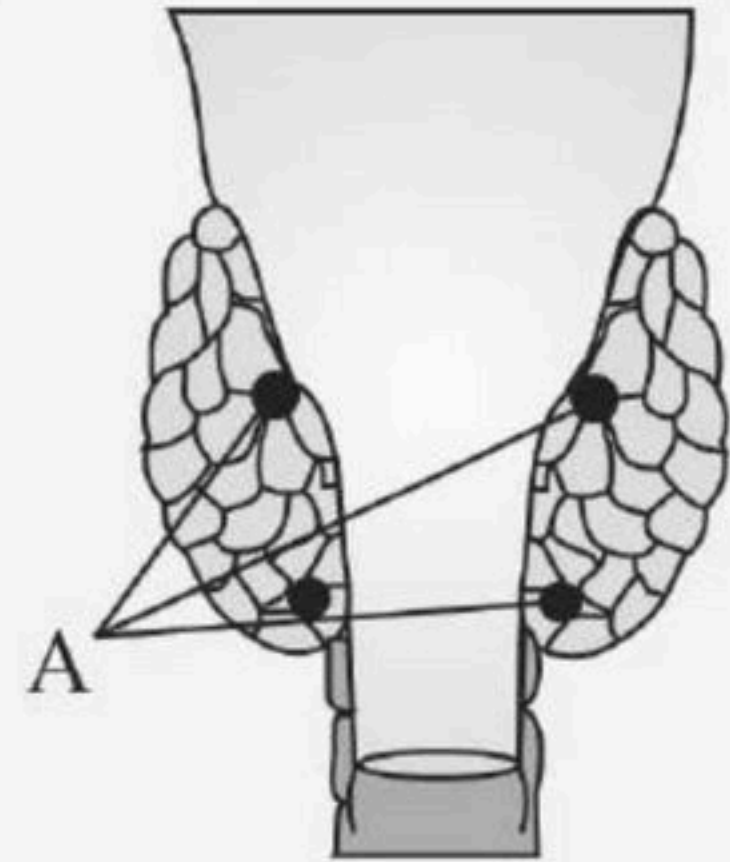
- ☒ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

66. The following figure shows view of the position of parathyroid gland. Identify the location of gland.

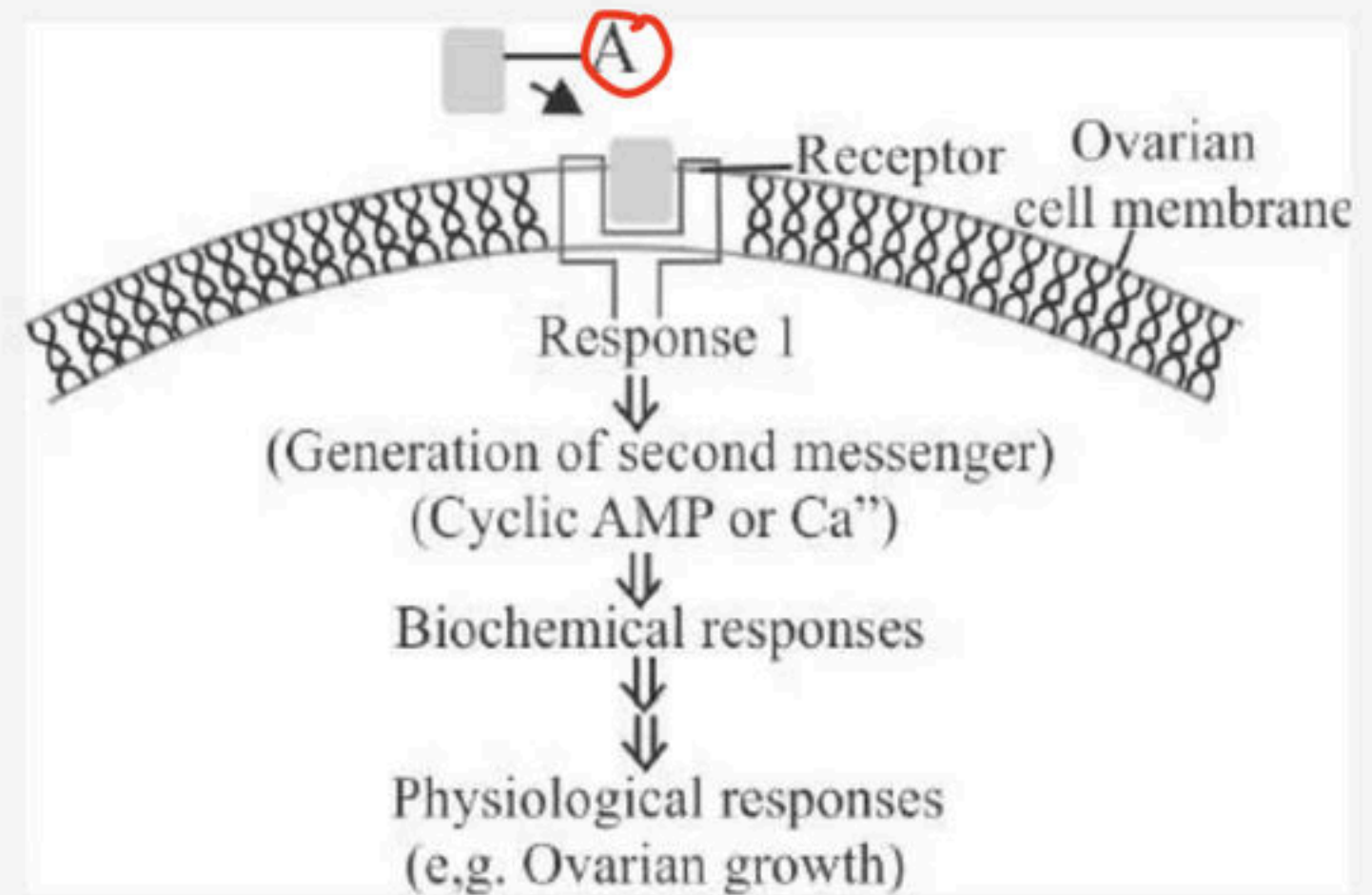
- (1) Dorsal side of forebrain
- ✓ (2) Present on the back side of the thyroid gland
- (3) Located on the anterior part of each kidney
- (4) Located on either side of the kidney



Title: Chemical Coordination and Integration

67. Following is the diagram of mechanism of hormone action. Identify A.

- (1) Estradiol
- (2) Testosterone
- ✓ (3) Follicle stimulating hormone
- (4) Cortisol



Title: Chemical Coordination and Integration

68. Statement A : Glucagon acts mainly on the hepatocytes and stimulates glycogenolysis resulting in an increased blood sugar.

2021

Statement B : Insulin acts mainly on hepatocytes and adipocytes and enhances cellular glucose uptake and utilisation.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

☒ (3) Both statements are correct.

(4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

69. Which of the following given features are appropriate for glucagon?

- (1) Peptide hormone which act on hepatocytes ✓
- (2) Increase the blood sugar by stimulating glycogenolysis ✓
- (3) Reduces the cellular glucose uptake and utilisation ✓
- (4) All of the above ✓✓



70. A child with following issues could have problems with which gland :-

1. A weak immune system

2. Short height

(1) 1. Thymus, 2. Thyroid gland only

(2) 1. Thymus, 2. Pituitary/Parathyroid gland

✓ (3) 1. Thymus, 2. Thyroid/ pituitary

(4) 1. Thymus, 2. Pituitary only



Title: Chemical Coordination and Integration

71. Identify the total number of peptide hormones from the following:

Erythropoietin, Thymosin, PTH, ANF, Gastrin, Secretin, ²⁰²⁵GIP, CCK, Insulin, Glucagon

(1) 8

(2) 7

(3) 9

(4) 10



Title: Chemical Coordination and Integration

72. Following (A to D) are the endocrine glands found in human body.

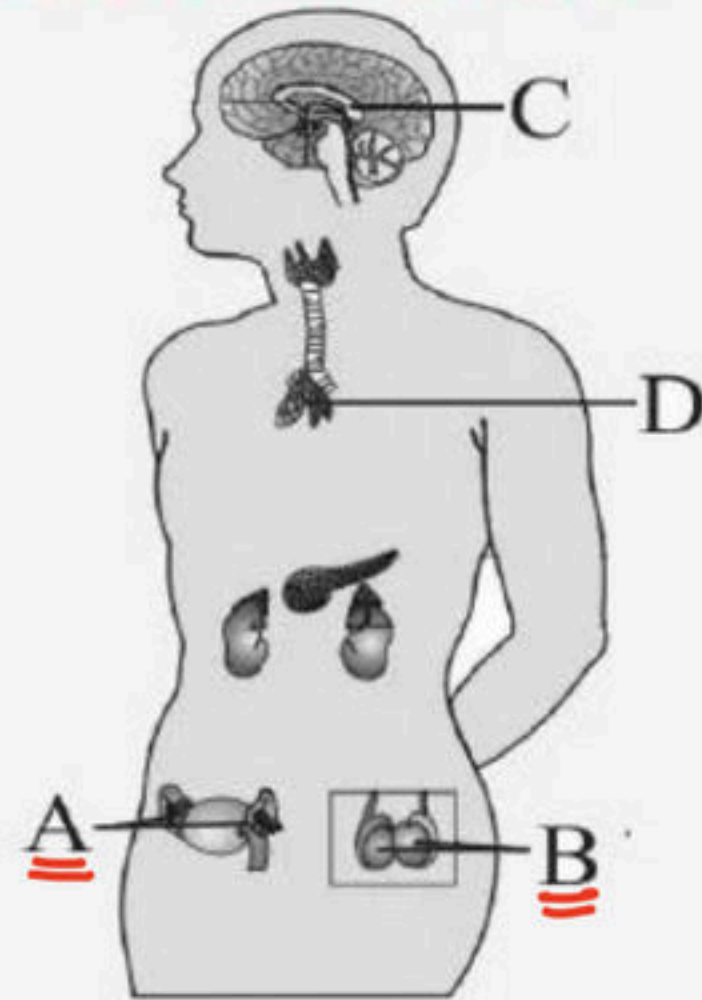
Identify the endocrine gland which is found in male and female respectively?

(1) C, D

(2) A, D

(3) B, C

☒ (4) B, A



Title: Chemical Coordination and Integration

73. PTH is a ____ hormone and its chemical nature is ____

- ✓ (1) Hypercalcemic, peptide
- (2) Hypocalcemic, peptide
- (3) Hypocalcemic, protein
- (4) Hypercalcemic, protein



74. Glucocorticoid causes all except

(1) Proteolysis

(2) Lipolysis

~~(3) Glycogenolysis~~ ✗

(4) Gluconeogenesis

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Title: Chemical Coordination and Integration

75. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Thyroid gland	P	Deficiency of iodine in our diet
B	Grave's disease	Q	Composed of follicles and stromal tissues
C	Hypothyroidism	R	Enlargement of thyroid gland

- (1) A - Q, B - R, C - P
(2) A - Q, B - P, C - R
(3) A - R, B - P, C - Q
(4) A - R, B - Q, C - P



Title: Chemical Coordination and Integration

76. The activity of formation of milk ____ and the ejection of milk is controlled by ____ respectively

2025

- (1) Oxytocin and oxytocin
- ✓(2) Prolactin and oxytocin
- (3) Prolactin and prolactin
- (4) Oxytocin and prolactin



77. Major role of thymus gland in human is/are

- (1) Differentiation of T-lymphocytes to provide cell mediated immunity
- (2) Differentiation of B-lymphocytes
- (3) Production of antibodies
- (4) 1 and 3



Title: Chemical Coordination and Integration

78. Statement A : A pair of ovaries located in the abdomen of female, which is the primary sex organ and produces one ovum during each menstrual cycle.

Statement B : Ovary is composed of ovarian follicles and stromal tissues, which produces steroid hormones called estrogen and progesterone.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

☒ (3) Both statements are correct.

(4) Both Statements are incorrect.



Title: Chemical Coordination and Integration

79. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Pancreas	P	Secretes glucagon
B	Islets of Langerhans	Q	Composite gland which acts as both exocrine and endocrine gland
C	Alpha cells	R	Secretes insulin
D	Beta cells	S	About 1-2 million in normal human pancreas

(1) A - S, B - P, C - Q, D - R

(2) A - P, B - S, C - Q, D - R

☒ (3) A - Q, B - S, C - P, D - R

(4) A - S, B - Q, C - P, D - R



80. Chemically epinephrine is

- ✓ (1) Amino acid derivative
- (2) Peptide hormone
- (3) Steroid hormone
- (4) Iodothyronines



Title: Chemical Coordination and Integration

81. Arrange in sequence, working of 'T₄'?

→ Intracellular receptor

1) Generation of second messenger ✗

2) Physiological response

3) Binding to membrane receptor

4) Biochemical response

(1) 1 → 2 → 3 → 4

(2) 3 → 1 → 4 → 2

(3) 4 → 3 → 2 → 1

(4) ✓ None of the above



Title: Chemical Coordination and Integration

82. Statement A : Endocrine glands lack ducts and are hence, called ductless glands. Their secretions are called hormones.

Statement B : Hormones are nutrient chemicals which act as intercellular messengers and are produced in trace amounts.

- ☒ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



83. Pancreas is a composite gland which acts as ____.

- (1) Both heterocrine and endocrine gland
- (2) Both exocrine and heterocrine gland
- ☒ (3) Both exocrine and endocrine gland
- (4) All of these



Title: Chemical Coordination and Integration

84. Statement A : The pituitary gland is located in a bony cavity called sella tursica and is attached to hypothalamus by a stalk.

Statement B : Pituitary gland is divided anatomically into an adenohypophysis and a neurohypophysis.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

☒ (3) Both statements are correct.

(4) Both Statements are incorrect.

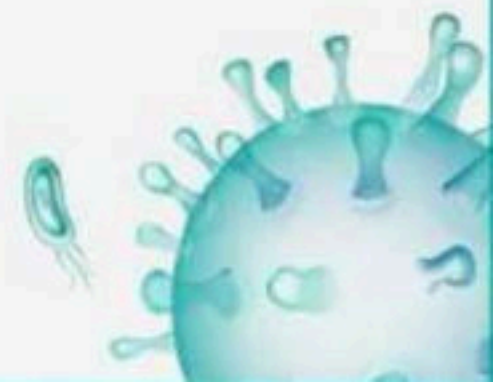


Title: Chemical Coordination and Integration

85. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Cholecystokinin	P	Secretes peptide hormone called erythropoietin
B	Gastric inhibitory peptide	Q	Stimulates secretion of pancreatic enzyme and bile juice
C	Growth factors	R	Inhibits gastric secretion and motility
D	Juxtaglomerular cells	S	Essential for normal growth of tissues and repairing

- (1) A - P, B - R, C - Q, D - S
(2) A - Q, B - P, C - R, D - S
☒ (3) A - Q, B - R, C - S, D - P
(4) A - S, B - R, C - Q, D - P



Title: Chemical Coordination and Integration

86. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Diabetes mellitus →	P	Deficiency of insulin
B	Diabetes insipidus	Q	Hyperthyroidism
C	Addison's Disease	R	ADH deficiency
D	Grave's Disease	S	Underproduction of hormones by adrenal cortex

- (1) A - Q, B - S, C - P, D - R
(2) A - Q, B - S, C - R, D - P
(3) A - S, B - Q, C - P, D - R
(4) ✓ A - P, B - R, C - S, D - Q



Title: Chemical Coordination and Integration

87. Match the hormones of Column I with its functions in Column II

Column I		Column II	
A	Catecholamines	P	Diurnal rhythm
B	MSH	Q	Immune response
C	Thymosins	R	Pigmentation
D	Melatonin	S	Stress hormone

- ✓ (1) A – S, B – R, C – Q, D – P
(2) A – R, B – Q, C – S, D – P
(3) A – Q, B – S, C – S, D – Q
(4) A – R, B – Q, C – S, D – S



Title: Chemical Coordination and Integration

88. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Testis	P	Produces testosterone
B	Leydig cells	Q	Ruptured follicle formed after ovulation
C	Ovaries	R	Composed of seminiferous tubules and stromal tissue
D	Corpus luteum	S	Composed of ovarian follicles & stromal tissues

- (1) A - R, B - P, C - S, D - Q
(2) A - Q, B - S, C - R, D - P
(3) A - R, B - P, C - Q, D - S
(4) A - Q, B - S, C - P, D - R



Title: Chemical Coordination and Integration

89. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Pineal gland →	P	Located on the dorsal side of forebrain
B	Melatonin	Q	Both lobes of thyroid gland are interconnected with a thin flap of connective tissue
C	Thyroid gland	R	Regulates sleep- wake cycle
D	Isthmus	S	Located on either side of the trachea

(1) A - P, B - Q, C - R, D - S

✓ (2) A - P, B - R, C - S, D - Q

(3) A - R, B - P, C - S, D - Q

(4) A - Q, B - R, C - S, D - P



Title: Chemical Coordination and Integration

90. Match the columns I and II and choose the correct option

Column I		Column II	
A	Thyrocalcitonin	P	Stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions
B	Cortisol	Q	Produces anti-inflammatory reactions and suppresses the immune response
C	Mineralo corticoids	R	Play role in the growth of axial hair, pubic hair and facial hair during puberty
D	Androgenic steroids	S	Decrease calcium levels in the blood

- (1) A - Q, B - S, C - P, D - R
(2) A - Q, B - S, C - R, D - P
☒ (3) A - S, B - Q, C - P, D - R
(4) A - P, B - R, C - S, D - Q



Title: Chemical Coordination and Integration

91. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Parathyroid gland	P	Result in the weak immune response in old persons
B	Thymosins	Q	Stimulates the process of bone resorption
C	T- lymphocytes	R	Also promotes <u>production of antibodies</u> to provide <u>humoral immunity</u>
D	Degeneration of thymus gland	S	Provide cell - mediated immunity

- (1) ✓ A - Q, B - R, C - S, D - P
(2) A - Q, B - R, C - P, D - S
(3) A - R, B - Q, C - S, D - P
(4) A - R, B - P, C - Q, D - S



Title: Chemical Coordination and Integration

92. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Cathecolamines	P	Helps in the maintenance of electrolytes , body fluidd volume and osmotic pressure
B	Glucocorticoids	Q	Involved in maintaining the cardio- vascular system
C	cortisol	R	Increases alertness , pupillary dilation , piloerection and sweating
D	Aldosterone	S	Stimulate gluconeogenesis, lipolysis and proteolysis

- (1) A - R, B - S, C - Q, D - P
(2) A - P, B - Q, C - R, D - S
(3) A - R, B - Q, C - S, D - P
(4) A - P, B - R, C - Q, D - S



Title: Chemical Coordination and Integration

93. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Membrane bound receptors	P	Present on <u>ovarian cell membrane</u>
B	Intracellular receptors	Q	Generate <u>secondary</u> messengers like Cyclic AMP
C	FSH receptors	R	Present in uterine cytoplasm
D	Estrogen receptors	S	Regulate gene expressions or chromosome function

- (1) A - P, B - S, C - Q, D - R
(2) A - Q, B - S, C - R, D - P
(3) A - S, B - Q, C - P, D - R
(4) A - P, B - R, C - S, D - Q



Title: Chemical Coordination and Integration

94. Select the right match of endocrine gland and their hormones among the options given below.

Column I		Column II	
A	Pineal	P	Epinephrine
B	Thyroid	Q	Melatonin
C	Ovary	R	Estrogen
D	Adrenal medulla	S	Tetraiodothyronine

- (1) A – S, B – Q, C – R, D – P
(2) A – Q, B – S, C – P, D – R
(3) A – S, B – Q, C – P, D – R
(4) A – Q, B – S, C – R, D – P



Title: Chemical Coordination and Integration

95. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Thymus	P	Produce a group of hormones called androgens
B	Thyroid gland	Q	Secretes peptide <u>hormone</u> called <u>thymosins</u>
C	Testis	R	Secretes steroid hormones called estrogen and progesterone
D	Ovary	S	Secretes a protein <u>hormone</u> called thyrocalcitonin

- ☒ (1) A - Q, B - S, C - P, D - R
(2) A - Q, B - S, C - R, D - P
(3) A - S, B - Q, C - P, D - R
(4) A - P, B - R, C - S, D - Q



Title: Chemical Coordination and Integration

96. Choose the correct option among the following

Column I		Column II	
A	Epinephrine	P	Stimulates muscle growth
B	Testosterone	Q	Decrease in blood pressure
C	Glucagon	R	Breakdown of liver glycogen
D	Atrial natriuretic	S	Increases heart beat factor

(1) A – Q, B – P, C – R, D – S

✓ (2) A – S, B – P, C – R, D – Q

(3) A – P, B – Q, C – R, D – S

(4) A – P, B – S, C – Q, D – R



Title: Chemical Coordination and Integration

97. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Hypothalamus	P	Under direct neural regulation of hypothalamus
B	Releasing hormones	Q	Inhibits the release of growth hormone from the pituitary gland
C	Somatostatin	R	Contains several groups of neuro - secretory cells called nuclei
D	Posterior pituitary hormones	S	Stimulate secretion of pituitary hormones

- ~~(1) A - R, B - S, C - Q, D - P~~
(2) A - S, B - R, C - Q, D - P
(3) A - Q, B - S, C - P, D - R
(4) A - P, B - Q, C - R, D - S



Title: Chemical Coordination and Integration

98. Column I contains some terms and column II contains their meanings. Match them properly and choose the right answer.

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Column I		Column II	
A	Glycogenesis	P	Conversion of glycogen to glucose
B	Glycosuria	Q	Conversion of <u>glucose</u> to <u>glycogen</u>
C	Glyconeogenesis	R	Excretion of <u>glucose in urine</u>
D	Glycogenolysis	S	Conversion of <u>non-carbohydrate</u> sources to <u>glucose</u>
		T	Conversion of glucose to starch

(1) A – P; B – Q; C – R; D – S

☒ (2) A – Q; B – R; C – S; D – P

(3) A – Q; B – P; C – R; D – S

(4) A – P; B – T; C – Q; D – S



Title: Chemical Coordination and Integration

99. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Testosterone	P	Secreted mainly by growing ovarian follicles
B	Estrogen	Q	Play major role in the process of spermatogenesis
C	Progesterone	R	Supports pregnancy

(1) A - R, B - P, C - Q

(2) A - R, B - Q, C - P

(3) A - Q, B - R, C - P

(4) A - Q, B - P, C - R



Title: Chemical Coordination and Integration

100. Match the columns I and II and choose the correct option.

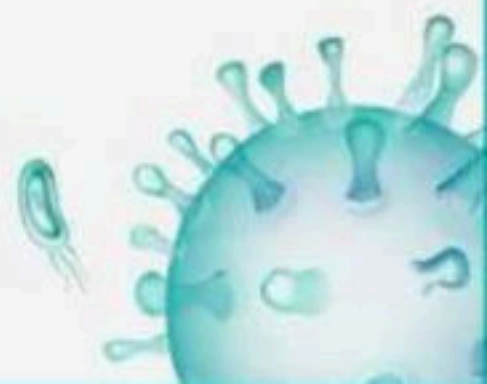
Column I		Column II	
A	Parathyroid gland	P	Lobular structure located on the ventral side of the heart and aorta
B	Parathormone	Q	Increases the Ca^{++} the blood
C	Thymus gland	R	Four glands are <u>present</u> on the backside of <u>thyroid gland</u>
D	Thymosins	S	Plays major role in the differentiation of T-lymphocytes

(1) A - S, B - Q, C - R, D - P

(2) A - R, B - P, C - Q, D - S

✓ (3) A - R, B - Q, C - P, D - S

(4) A - P, B - Q, C - R, D - P



THANK YOU



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Thank You

